

Exercise 1: Navigation and Title Verification Write a script that launches a browser and navigates to the Amazon home page. Once loaded, capture the page title and use an assertion to verify that it contains the word "Amazon". After the check is complete, navigate to the "Mobiles" category page and go back to the home page using browser navigation methods.

```
from selenium import webdriver

from selenium.webdriver.common.by import By
import time

# Launch browser
driver = webdriver.Chrome()

# Open Amazon home page
driver.get("https://www.amazon.in")

# Maximize window
driver.maximize_window()

# Get page title
title = driver.title

# Print title
print("Page Title:", title)

# Verify title contains Amazon
assert "Amazon" in title

# Open Mobiles page
mobiles_link = driver.find_element(By.LINK_TEXT, "Mobiles")
mobiles_link.click()

# Wait for page load
time.sleep(3)
```

```
# Go back to home page
```

```
driver.back()
```

```
# Wait again
```

```
time.sleep(3)
```

```
# Close browser
```

```
driver.quit()
```

Exercise 2: Basic Locators and Search Identify the Amazon search bar using its `id` and the search button using its `xpath`. Type "Wireless Headphones" into the search bar and click the search button. Verify that the results page displayed contains a specific header or text indicating that results for your search are being shown.

```
from selenium import webdriver
```

```
from selenium.webdriver.common.by import By
```

```
import time
```

```
# Launch browser
```

```
driver = webdriver.Chrome()
```

```
# Open Amazon website
```

```
driver.get("https://www.amazon.in")
```

```
# Maximize browser window
```

```
driver.maximize_window()
```

```
# Locate search bar using id
```

```
search_box = driver.find_element(By.ID, "twotabsearchtextbox")
```

```
# Enter product name
```

```
search_box.send_keys("Wireless Headphones")

# Locate search button using xpath

search_button = driver.find_element(By.XPATH, "//input[@id='nav-search-submit-button']")

# Click search button

search_button.click()

# Wait for results page to load

time.sleep(3)

# Verify results page contains searched text

page_source = driver.page_source

assert "Wireless Headphones" in page_source

print("Search results displayed successfully")

# Close browser

driver.quit()
```

Exercise 3: Implementing Implicit and Explicit Waits Create a test that searches for a specific laptop model. Implement an Implicit Wait at the driver level to handle general loading. Then, use an Explicit Wait (`WebDriverWait`) with an `ExpectedCondition` to wait specifically for the "Results" grid or a specific product image to become visible before attempting to click the first result.

```
from selenium import webdriver

from selenium.webdriver.common.by import By

from selenium.webdriver.common.keys import Keys

from selenium.webdriver.support.ui import WebDriverWait

from selenium.webdriver.support import expected_conditions as EC

import time
```

```
# Launch browser
driver = webdriver.Chrome()

# Maximize window
driver.maximize_window()

# Implicit Wait
driver.implicitly_wait(10)

# Open Amazon
driver.get("https://www.amazon.in")

# Search for laptop model
search_box = driver.find_element(By.ID, "twotabsearchtextbox")
search_box.send_keys("Dell Inspiron Laptop")
search_box.send_keys(Keys.RETURN)


# Explicit Wait for first product image/result
wait = WebDriverWait(driver, 10)

first_product = wait.until(
    EC.visibility_of_element_located(
        (By.XPATH, "//img[contains(@class,'s-image')]")
    )
)

# Click first result
first_product.click()

# Wait for few seconds
time.sleep(3)

# Close browser
```

```
driver.quit()
```

Exercise 4: Advanced Locators (CSS Selectors & Links) Navigate to the Amazon footer section. Use CSS Selectors to identify and click on the "About Us" link. On the following page, find a specific element using its `name` or `link\_text` and print its text content to the console.

```
from selenium import webdriver
```

```
from selenium.webdriver.common.by import By
```

```
import time
```

```
# Launch Chrome browser
```

```
driver = webdriver.Chrome()
```

```
# Open Amazon website
```

```
driver.get("https://www.amazon.in")
```

```
# Maximize browser window
```

```
driver.maximize_window()
```

```
# Scroll to footer section
```

```
driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")
```

```
# Wait for page loading
```

```
time.sleep(3)
```

```
# Locate and click "About Us" using CSS Selector
```

```
about_us = driver.find_element(
```

```
    By.CSS_SELECTOR,
```

```
    "a[href*='aboutamazon']")
```

)

```
about_us.click()
# Wait for next page
time.sleep(3)
# Find element using LINK_TEXT
element = driver.find_element(By.LINK_TEXT, "Amazon Science")
# Print text content
print("Text Content:", element.text)
# Close browser
driver.quit()
```

Exercise 5: Element Interaction and Synchronization Automate the process of filtering search results. Search for "Smart Watches," then locate the "Brand" filter in the left-hand sidebar. Click on a specific brand checkbox (e.g., "Samsung" or "Apple"). Because filters refresh the page asynchronously, implement a wait strategy to ensure the page has updated before counting how many products are displayed on the first page.

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

# Launch browser
```

```
driver = webdriver.Chrome()
# Maximize window
driver.maximize_window()
# Open Amazon website
driver.get("https://www.amazon.in")
# Implicit wait
driver.implicitly_wait(10)
# Search for Smart Watches
search_box = driver.find_element(By.ID, "twotabsearchtextbox")
search_box.send_keys("Smart Watches")
search_box.send_keys(Keys.RETURN)
# Explicit wait for filter section
wait = WebDriverWait(driver, 15)
# Click Samsung brand checkbox
brand_filter = wait.until(
    EC.element_to_be_clickable(
        (By.XPATH, "//span[text()='Samsung']")
    )
)

brand_filter.click()

# Wait for page refresh after applying filter
wait.until(
    EC.visibility_of_element_located(
        (By.XPATH, "//div[@data-component-type='s-search-result']")
    )
)
```

```
)  
)  
  
# Get all products displayed on first page  
products = driver.find_elements(  
    By.XPATH,  
    "//div[@data-component-type='s-search-result']"  
)  
  
# Print product count  
print("Number of products displayed:", len(products))  
# Wait for few seconds  
time.sleep(3)  
# Close browser  
driver.quit()
```